

An Exploration of Default Images in Text-to-Image Generation

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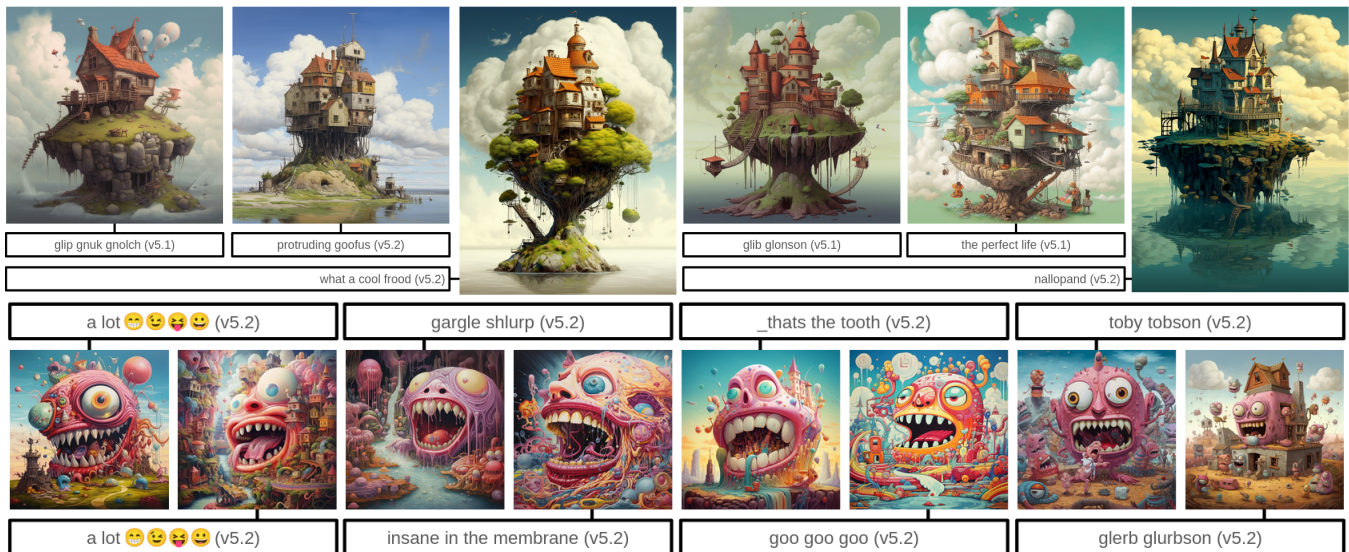


Figure 1: Default images in Midjourney. Varied prompts lead to visually similar outputs, motivating our exploration of default image behavior.

Abstract

In the creative practice of text-to-image (TTI) generation, images are synthesized from textual prompts. By design, TTI models always yield an output, even if the prompt contains unknown terms. In this case, the model may generate default images: images that closely resemble each other across many unrelated prompts. Studying default images is valuable for designing better solutions for prompt engineering and TTI generation. We present the first investigation into default images on Midjourney. We describe an initial study in

which we manually created input prompts triggering default images, and several ablation studies. Building on these, we conduct a computational analysis of over 750,000 images, revealing consistent default images across unrelated prompts. We also conduct an online user study investigating how default images may affect user satisfaction. Our work lays the foundation for understanding default images in TTI generation, highlighting their practical relevance as well as challenges and future research directions.



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CCS Concepts

• Human-centered computing → Empirical studies in HCI; • Computing methodologies → Neural networks; • Applied computing → Media arts.

Keywords

text-to-image generation, Midjourney, AI art, prompt engineering, creativity

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1 Introduction

Generative artificial intelligence (AI) has swept into mainstream. One popular type of generative AI is text-to-image (TTI) generation, where images are created by prompting the generative model with text [16, 61, 62]. The prompt consists of a textual description of the envisioned image, including keywords that are known to trigger specific subjects, forms, and styles [45, 60, 64]. The practice of ‘prompt engineering’ is key to TTI generation [61]. Practitioners craft a prompt, observe the outcome, and adjust the prompt until a satisfactory outcome is achieved [62].

While there is a growing number of studies investigating the practice of prompt engineering or prompt design for TTI generation (e.g., [7, 45, 50, 52, 62, 65, 83]), and the design of user interfaces for prompt engineering (e.g., [2, 26, 40, 51]), less attention has been given to how models behave when prompts are misunderstood or ambiguously interpreted. TTI models are trained to always yield an output, even if the input prompt contains terms and concepts not known to the model. When such a prompt is given to the TTI model, it may generate what we call *default images* – visually similar images resulting from dissimilar prompts (see figures 1 and 5, and Appendix 10). These images are seemingly unrelated to the input prompt and, therefore, often do not match the user’s expectations.

Default images reveal gaps in the generative model’s vocabulary and capabilities. For instance, a creative practitioner may prompt a TTI model with the Tagalog word “*pagpag*” (referring to a dish made from leftover food scavenged from garbage dumps), resulting in a default image (e.g., a painting of a woman with a cloud covering her face). This indicates that the prompt was not understood and the practitioner should revise the wording to improve the generated image. Default images can put an early end to a potentially unfruitful pathway in the practitioner’s prompting of the TTI model, but they can also derail the entire creative process, leading the user to abandon the quest to generate their desired image (cf. Section 2.2). Therefore, default images and their recognition play a crucial role in the creative practice of prompt engineering and for discovering the limits of TTI models and their (computational) creativity. Understanding how generative systems respond to unfamiliar or ill-formed prompts is essential for designing interactive systems that support creative exploration and for providing feedback to users. In human–AI co-creation, unexpected outputs (such as default images) can disrupt the feedback loop between user intent and system output, introducing ambiguity or undifferentiation. This undermines trust, perceived accuracy, usability, and user agency [24, 33, 72]. The misalignment between user intent and outputs can lead to frustration and a sense of lost control over the creative process [48]. Investigating the phenomenon of default images contributes to

broader HCI discussions on transparency, interpretability, and the design of meaningful interactions with generative models. Default images have not received attention in prior research.

In this paper, we present the first exploration of default images in Midjourney,¹ a popular image generation system. We describe our systematic approach to manually create input prompts likely triggering default images and our initial experiments in generating images from these prompts. We then apply affinity diagramming to find similarities between the generated images and identify ten default images. We further conduct an online user study investigating how default images impact user satisfaction, and how they align with human perception and expectations. Scaling up our analysis, we then present insights from a large-scale computational analysis of images on Midjourney.

We make the following contributions:

- *Conceptual and empirical contribution:* We define the concept of *default images* and provide examples of this phenomenon. Based on an initial study, we contribute six categories of prompts that are likely to elicit default images. We present the results of our manual experiments with these prompts and several ablation studies. We also present a user study with crowdsourced participants (N = 48) investigating how default images may affect user satisfaction.
- *Methodological contribution:* We describe our initial systematic approach for manually creating prompts that trigger default images. We then describe a simple method for identifying default images at scale, first clustering images by visual similarity and then filtering by intra-cluster prompt diversity. With this method, we computationally analyze over 750,000 images from Midjourney, providing empirical evidence of the existence of default images in response to user-written prompts on Midjourney.
- *Open-source contribution:* We share a dataset with 189,432 prompts and 757,728 candidates of default images, enabling future research on default images (see <https://huggingface.co/datasets/tti-dev/default-images>).

We discuss the challenges of identifying unique but realistic input prompts that reliably trigger default images, and the challenges of identifying default images in a real-world setting. Based on the results of our investigation, we make eight postulates, detailing the characteristics of default images in generative TTI models and providing a basis for future experimentation and theoretical development.

2 Default Images in Text-to-Image Generation

2.1 What are Default Images?

Some inputs to TTI generation systems may seemingly fail to generate the intended results from the creative practitioner’s perspective. We call this failure mode *default images*. We use the term ‘default’ to denote that TTI models resort to generating these images when they do not recognize their input. This could be because the input terms were not part of the generative model’s training dataset or because the terms in the prompt do not invoke strong imagery. In these cases, distinct prompts may yield near-identical images. In

¹www.midjourney.com

machine learning, a related concept is called mode collapse [19], a failure that limits the usefulness of generative models for image synthesis.

The defining characteristic of default images is that they are encountered in response to multiple different and unrelated prompts. We demonstrate this in Figure 1 and Figure 5. While there are often slight variations to the default images, they can be remarkably stable in terms of motif (i.e., subject) and style (cf. Figure 5), making default images distinctly recognizable. Experienced practitioners and in particular creative practitioners who seek to deviate from common prompts will likely have encountered default images in their prompting. This paradoxical situation, which may go unnoticed by most users, reflects the current limitations of TTI models. As Manovich observes, AI image generators “deal well with frequently occurring items and patterns in the data but do not know what to do with the infrequent and uncommon” [54]. These models “are much better at generating visuals of popular subjects and aesthetics than unfamiliar or rare ones.”

Artists and other creative practitioners often seek to generate images that are novel [32], a key component of creativity [75]. Default images may pose a particular challenge to these users, since default images with their generic visuals tend to occur with uncommon and highly original prompts. Default images may satisfy users with less specific image requirements or lower task persistence. In these cases, users may engage in cognitive gap-filling processes [93] or construct personal associations [5], resulting in images that appear more relevant to the prompts than they actually are. This effect may be further amplified when the generated images have strong aesthetic appeal, since aesthetically pleasing stimuli are often judged more favorably [70] and attractive designs are often perceived as more effective, regardless of their actual function [35]. Users may accept visually appealing default outputs rather than investing effort to generate better alternatives. It is detrimental to creativity overall if TTI users are pushed towards repetitive, generic images in response to their novel prompting. This issue makes the phenomenon of default images particularly relevant to visual creativity and originality.

Default images primarily result from how text-to-image models structure internal representations of visual concepts. When presented with ambiguous or unknown inputs, these models rely on their learned latent space – the continuous, high-dimensional internal space encoding visual relationships. Regions of this space correspond to generic or frequently occurring visual patterns, which the model defaults to in the absence of specific semantic guidance. Stochastic sampling adds variability, and over-training or memorization of frequent training examples can reinforce these biases. Ultimately, default images arise from latent space’s inherent structure and learned biases.

We can use a metaphor to help understand default images: the latent space as a vast landscape. Images are generated by sampling from regions within this landscape. Known concepts are easy to spot (think of them as mountains and hills that attract attention). But there are also planes that connect vast regions of the space, which do not correspond strongly to specific learned concepts. Default images result when the TTI model samples from these planes because of the absence of strong ‘attractors’ (known concepts) in the sampled region of latent space.

2.2 Importance of Studying Default Images

Many Midjourney users type long, descriptive prompts containing specific keywords, representing known concepts and style modifiers [60, 62]. Known concepts reliably elicit the desired imagery, but they also act like ‘magnets’ that overshadow any unknown concepts, moving the output away from default images towards imagery related to the known concept. For instance, including “trump” or even “donald” in a Midjourney prompt may invoke images of a certain U.S. president in many different contexts. This means the majority of Midjourney users will likely not encounter default images in their image creation efforts. So why study default images?

Default images provide valuable insights into the robustness and limitations of TTI models. When models generate similar outputs from diverse, unrelated prompts, this indicates semantic gaps and challenges with linguistic constructs, niche topics, or abstract concepts, and a lack of generalization (i.e., the ability of a model to perform well on unseen data). Such patterns suggest that the model may rely on memorized outputs rather than generating novel imagery, revealing limits in the model’s computational creativity. Awareness of these issues can help with assessing model capabilities, benchmarking TTI model performance, and detecting vulnerabilities related to adversarial prompting.

Understanding default images informs data and model quality improvements. Default images can indicate that the training data lacks diversity or contains biased patterns, for instance, highlighting insufficient coverage of low-resource languages, such as Tagalog, an Austronesian language spoken as a first language by about a quarter of the population of the Philippines. Addressing these cases can enhance dataset curation and fine-tuning, promoting more inclusive and culturally diverse images. Studying this phenomenon can help monitor the risk of default images entering training datasets, which could amplify the self-reinforcing loop of AI-generated media [3, 25, 99].

Understanding default images is valuable for creative practitioners and prompt engineers exploring computational creativity and comparing different generative models. Default images reveal the model’s limitations in interpreting abstract or unfamiliar concepts. Knowledge of these gaps allows users to refine prompts and circumvent “dead zones” where outputs are generic or repetitive, and implement strategies to recover from failures, ultimately supporting more meaningful and original visual outputs.

Default images also have potential security implications. An attacker could infer from the appearance of default images the model’s limitations or gaps in training data. This knowledge could be exploited to reverse-engineer aspects of the model’s dataset or architecture. Default images can act as a model signature, allowing attackers to craft prompts that identify the TTI model or version being used. This information could be help launch more targeted attacks. For instance, knowing that a model produces a predictable default image for certain prompts could allow attackers to inject misleading or confusing input in systems that rely on generative models.

From a user experience perspective, recognizing patterns in default image generation can inform more resilient and user-friendly

interface design. When users encounter similar outputs from diverse prompts, it can be frustrating. Interfaces that provide feedback, suggest alternative prompts, or guide users through effective prompt engineering can help navigate these limitations and recover from failures, enhancing creativity and usability.

3 Related Work

Our review of related work covers three intertwined topics: 1) how TTI models are built and prompted, 2) known failure modes and “degenerate” outputs in generative systems, and 3) user-facing evaluations of generation quality and satisfaction.

3.1 Text-to-Image Generation and Prompt Engineering

Text-to-image generation has emerged as a powerful subfield of generative AI, in which deep neural networks (e.g., text-conditioned diffusion models) learn to translate free-form textual prompts consisting of discrete words into high-fidelity images. Prominent TTI systems, such as Midjourney, OpenAI’s DALL-E, and Stable Diffusion, demonstrate the ability to produce photorealistic or artistically stylized images across many domains. Generative TTI models are pre-trained on extensive datasets and generate images for a given prompt by probabilistic sampling from their latent space.

While TTI systems attempt to match human expectations, the prompting language is loose and it is near impossible to describe every detail of an envisioned output in one textual prompt, with a limited number of characters [17]. Further, the model may lack relevant prior knowledge and context. While special prompt keywords can address some of these shortcomings, such as improving the resulting image quality and imbuing it with certain styles [60], TTI models have specific failure modes that we review in the following section.

3.2 Failure Modes in Generative Models

TTI outputs frequently diverge from user intent due to the limitations of prompting as an interface [57]. One reason is semantic ambiguity. Polysemous terms (such as “bank”, “bat”, or “ball”) can yield images that reflect unintended word senses, creating a mismatch between the user’s mental model and the generative model’s output [69, 94]. Users may intentionally use polysemous and ambiguous terms in their creative practice. So-called “magic terms” – lexical additives intentionally designed to induce randomness or probe latent failure boundaries – may result in unpredictable or uncanny imagery [53, 60, 66].

Liu and Chilton presented a manual analysis of failure modes in text-to-image models [45], identifying recurring breakdowns including underspecified prompts, overloaded terms, and model over-reliance on stereotypical associations. Their findings highlight that many failures stem from subtle mismatches between user phrasing and model priors. This research was later extended to image prompts. Qiao et al. categorized sources of model misinterpretation and proposed design strategies to expose latent failure cases [65]. These studies contribute to a broader understanding of how prompt formulation interacts with failure patterns in TTI systems. Failure modes are relevant in interfaces for reviewing and auditing TTI models [53, 67]. Quaye et al. presented Adversarial

Nibbler, a system for crowdsourcing the collection of adversarial examples [66], and Maldaner et al. presented Mirage, a system for crowdsourced auditing of generative models [12, 53].

Beyond adversarial prompts, training artifacts can manifest as bizarre text–image associations. As these pairs have been drawn from the internet, they can include incorrect assignments or mismatched correspondences [16]. Daras and Dimakis found “strange” associations between special terms in prompts and concepts in generated images [11]. They speculated that these associations in TTI models constituted a “secret language”. Williams et al. investigated the effects of non-mainstream dialects in TTI [95], finding that minimal changes in prompts (including dialect) can manifest biases in TTI models. Community reports of special tokens in language models (and by extension, in TTI models), such as “*SolidGoldMagikarp*”, highlight spontaneous ‘glitch’ phenomena that can arise without adversarial intent [42, 74].

In addition to semantic and training-related failures, TTI models exhibit high stochasticity, introducing a form of variability that itself constitutes a failure mode. Identical prompts can produce substantially different images across runs, complicating both user expectation and scientific reproducibility. Strategies such as sampling multiple random seeds and aggregating outputs have been proposed to mitigate this variability [46], and studies show that seed choice alone can influence the emergence of rare or novel concepts [78]. Failures of prompt recognition – where the model fails to parse novel or unsupported inputs and defaults to placeholder imagery – remain underexamined. Our work targets this gap by presenting a systematic approach for eliciting, identifying, and classifying default-image failures in Midjourney.

3.3 Detecting Mode Collapse in Text-to-Image Models

Mode collapse is a failure mode in generative models where the model produces less diverse outputs than expected. It was first noted by Goodfellow et al. in Generative Adversarial Networks (GANs) [19]. Default images share characteristics with mode collapse: both involve gravitating toward a limited set of outputs and occur only in response to specific prompts.

Various techniques for detecting and mitigating mode collapse in TTI models have been developed and applied to GANs and diffusion models. Barsha and Eberle studied early detection of mode collapse through loss monitoring [4]. Other techniques include multi-generator approaches (e.g., MTC-GAN [101]), regularization techniques (e.g., ICCR [84]), contrastive learning (e.g., MCL [20]), and novel architectures (e.g., style-guided discriminator [10]). Bau et al. used semantic segmentation networks to compare the distribution of objects in generated images with the target distribution to identify omitted classes [6]. Hashemi et al. developed Connective Novelty Detection, a method that uses an autoencoder and a binary classifier to detect out-of-distribution samples. By creating negative instances close to the positive class distribution and combining real and generated samples, the method improves the robustness of detecting mode collapse [22]. Gong et al. used Feature Distribution Divergence Minimization to minimize the divergence between the distributions of real and generated features, reducing the learning pressure on the discriminator and addressing mode collapse

without requiring prior knowledge or manual design [18]. Duym et al. developed the Mode Collapse Entropy (MCE) score [14] to overcome weaknesses of previous metrics in real-world scenarios.

These approaches are applied at training time and assume full access to either the model or training data, or both, making them impracticable for use in black-box models, such as Midjourney, where the training data is not known and the model is not directly accessible. This is particularly important in the context of TTI generation, as the opacity of training image datasets raises questions concerning data provenance, diversity, quality, and representativeness.

Several inference-time methods have been proposed for detecting mode collapse in black-box models. Statistical, sampling-based tools can be used to visualize and quantify intra-mode collapse without access to training data or model parameters, providing a practical approach for black-box diagnosis and calibration [97]. Inversion-based inference perturbation (IIP) alters the inference process to detect memorized images, using unconditional DDIM inversion to extract latent codes and identify images exhibiting collapse-like behavior [29]. EncoderMI leverages membership inference against contrastive learning-based image encoders, exploiting their overfitting tendencies to reveal whether inputs belong to the training set, which can indirectly indicate mode collapse [44]. Binary noise guidance learning (BnGLGAN) introduces distribution-guided noise reconstruction to stabilize image translation and mitigate collapse [100].

While these techniques require careful statistical modeling or inference perturbation, we adopt a simpler approach. We exploit the properties of default images – visually similar outputs generated from dissimilar prompts – to identify mode collapse in black-box TTI models (Section 4.5).

3.4 Experiments and User Studies on TTI Quality and Satisfaction

Evaluations of TTI models often rely on automatic metrics and curated benchmarks, yet these measures are known to misalign with human perception [36, 63]. The Fréchet Inception Distance (FID), for instance, can yield scores that disagree with user judgments of image quality and prompt alignment, limiting its utility as a sole evaluation criterion [21, 36, 63]. Complementary metrics – such as CLIPScore [23] – have been proposed to capture semantic fidelity, but also exhibit gaps when compared to subjective assessments [43]. Recognizing these shortcomings, Russo suggested criteria for a creative TTI benchmark incorporating insights from linguistics, cognitive psychology, and art theory to better capture human notions of creativity and relevance [76]. More recently, the Holistic Evaluation of Text-to-Image Models (HEIM) benchmark systematically assessed 26 state-of-the-art TTI generators across twelve dimensions (ranging from aesthetics and originality to bias and robustness) using a combination of automated measures and human ratings, providing a richer, multidimensional view of model performance [41].

Human-centered studies bridge the gap between automatic scores and actual user experience. The Pick-a-Pic dataset includes over half a million human preference annotations, enabling the training of PickScore, a model predicting human rankings of generated

images with superhuman accuracy, and suggesting a more reliable protocol for evaluating TTI outputs [31]. Researchers extended this work with ImageReward, a preference-based reward model from 137,000 expert comparisons. Fine-tuning diffusion models against this reward function produces images that users consistently rate as higher quality [98]. Otani et al. highlight the lack of verifiable, reproducible human evaluation in TTI research and propose a standardized, crowdsourced protocol [63]. Their results reveal that automatic metrics frequently diverge from human judgments, underscoring the need for rigorous user studies. In interactive settings, tools, such as PromptAid [56] and PromptCharm [89], helped participants craft prompts yielding images better aligned with their intent, with users reporting increased satisfaction and perceived control over output quality. Multi-turn guidance systems, such as DialPrompt [47], achieve significant gains in both image quality ratings and user-centricity scores compared to baseline prompt-engineering interfaces. Finally, exploratory work in teacher education demonstrates how practitioners reflect on prompt strategies and generator behaviors in art and design, revealing both learning affordances and cognitive challenges when integrating TTI tools into creative pedagogy [71].

Together, this prior work illustrates the complex interplay between objective metrics, human preference, and interaction design in assessing and improving TTI systems.

4 Method

We present the first investigation into default images, acknowledging that the complexity of TTI models limits us to a tiny glimpse of the latent space and its default images. Our approach involves systematic prompt creation (Section 4.1), manual image generation experiments and ablation studies to probe the latent space (Section 4.2), identification of default images using affinity diagramming (Section 4.3), a user study on user satisfaction with default images (Section 4.4), and a large-scale computational analysis of Midjourney images (Section 4.5).

4.1 Input Prompts Likely Triggering Default Images

In our initial exploration of the phenomenon of default images, we explored three categories of prompts that could result in a conceptual mismatch between user expectations and the resulting image. We first tried to ‘confuse’ the model with polysemous words or prompts outside the training data (e.g., e-mail addresses), but the rate of default images was too low. We then tried prompts that were likely not recognized by the model and lacked practical purpose, including (but not limited to): contextually misleading phrases (e.g., *Einstein’s relativity theology*), random character strings, emojis, and special characters (e.g., *aölkjnewjbtn* and *%*), lyrics and misheard lyrics, and abstract and metaphorical concepts that may not invoke strong imagery (e.g., *absolute ownership* and *extreme aptitude*). We excluded prompts in this category from further analysis because, although they often trigger default images, they are not intended to produce coherent visual content. These prompts often function as stress tests or purposeful creative experiments rather than artistic image generation attempts. While these prompts remain important for future investigations into model boundaries and creative

prompting strategies, in this work, we focus on default images arising from realistic prompting scenarios, where users aim to generate meaningful images (instead of default images).

We explored prompts that involve both unrecognized terms and practical use cases, a combination likely found in real world use, but that still challenge the TTI model. These include:

- A1: Rare or uncommon names, such as fantasy names, (e.g., *Kirk Kirkson* and *Shelan Creswell*)
- A2: Corrupted words or altered words (e.g., *tractoajrte*)
- A3: Web addresses (e.g., *www.facebook.com* and *www.quora.com*)
- A4: Words from low-resource languages, like A4.1 Finnish (e.g., *hevonen* [horse]), and A4.2 Tagalog, an Austronesian language spoken mainly in the Philippines, (e.g., *gubat* [forest])
- A5: Glitch tokens [74] (e.g., *rawdownload*)
- A6: Syllables or abbreviations (e.g., *BCD* and *YMCA*)

We manually created a set of 130 prompts based on the six categories, A1–A6. We created prompts in category A2 by introducing altering known words with slight variations. We selected the two low-resource languages (A4.1 Finnish and A4.2 Tagalog) for their language-native words that are not borrowed from other languages, so they do not cognate with widely known foreign languages. We collected the glitch tokens from related literature [74, 91, 92]. The prompts are in Appendix A).

To assess the likelihood that the selected prompts had been encountered during model training, we checked each prompt on haveibeentrained.com, a search interface for over 5.8 billion image-caption pairs. The tool helps users determine whether a specific term (e.g., their name) appears in large-scale datasets, such as LAION [38]. These datasets are commonly used in the TTI model training [38, 80]. We sought prompts that were absent from – or underrepresented in – the training data, increasing the likelihood that the resulting generated imagery would involve default or fallback behavior. Our inclusion criteria were twofold: 1) the prompt should plausibly occur in real-world use, and 2) it should pose a challenge to the model due to low training data coverage. We describe our image generation process next.

4.2 Image Generation Experiments and Ablation Studies

We used Midjourney’s webpage² (see Figure 2) to execute the experiments, using fixed parameters listed in Table 1. For each of the 130 prompts in the six input categories (A1–A6), we ran 1 trial, resulting in four images per prompt, for a total of 520 images. We then visually analyzed the images, as described in Section 4.2.

Table 1: Image generation parameters used in manual study

Parameter	Value	Description
model	6.1	Setting the model version
aspect ratio	1	Standardizing the image format to square
stylize	0	Controlling the artistic flair
chaos	0	Controlling surprising random outputs
weird	0	Controlling quirky unconventional outputs
seed value	123456	Ensuring consistency and reproducibility

²www.midjourney.com/imagine

As a sensitivity analysis to test the robustness of default images, we conducted five small-scale ablation studies. We investigated the effect of different seed values on the default image (Section 5.2.1). We explored whether default images could be altered with prompt modifiers [60] (Section 5.2.2). We then examined default image behavior within larger prompts, including known terms (Section 5.2.3) and combinations of prompts known to trigger default images (Section 5.2.4). In our last study, we investigated default images in different model versions (Section 5.2.5).

4.3 Default Image Identification

To identify default images in the set of generated images, we used affinity diagramming (or KJ Method) [81]. It is a qualitative method that organizes data into groups based on perceived similarity. The method is commonly employed to categorize complex data. The first two authors collaboratively grouped images on a digital canvas based on their perceived similarity. During the session, the two authors discussed their rationale and iteratively refined the groups. Simultaneously, the two authors developed descriptive labels for the images (e.g., “Lady-Birdhead”), enabling them to refer to these images by name. They continued this collaborative process until all images were either categorized as default images (those with strong similarities across multiple images created with different prompts) or not. Identifying similarities between images was often straightforward despite subtle variations, as visible in Figure 5.

4.4 Online User Study

We used crowdsourcing through Prolific,³ an online human subject pool, to explore participant perception of default images, specifically, the impact of default images on user satisfaction. We used an online questionnaire presented participants with four images, including the original prompt that created the image. Unknown to the participant, for two of the images (Q1 and Q2), we replaced one term in the prompt with a term identified as causing default images. For instance, in Q1, we replaced ‘raven’ in the prompt “*photoshoot of a woman wearing a white off-shoulder dress with a black raven on top of her head*” with ‘Greagoft’. For Q2, we replaced “leaves and flowers” with “Shuttastfr”. With this replacement, we simulated the situation where a word is unknown to the generative model. We showed participants the original prompt and the image resulting from the modified prompt, simulating the mismatch in expectations that an encounter with a default image can produce. For Q3, we used an image that matched the descriptive prompt, without replacements. For Q4, we used a prompt that was unknown to the model and, thus, the accompanying image was a default image. The four prompts and images are in Figure 8. We asked participants to rate their satisfaction with each image, given the prompt, on a scale from 1 (Strongly Dissatisfied) to 7 (Strongly Satisfied). Participants received £6.29 per hour, and the median time to complete the survey was 3 min and 49 seconds.

³www.prolific.com

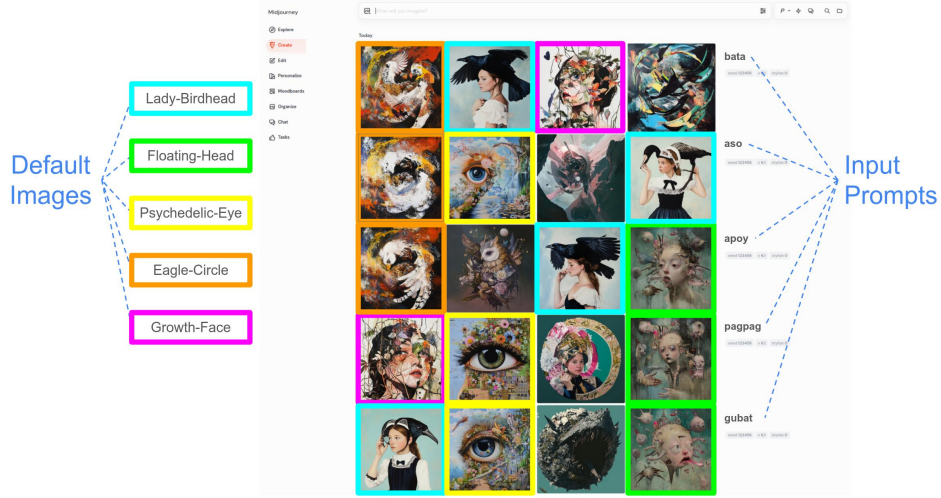


Figure 2: Default-prompt relationships in Midjourney. A screenshot illustrating how diverse input prompts (e.g., the Tagalog words *bata*, *aso*, *apoy*) yield overlapping clusters of recurring default images.

4.5 Computational Analysis of Midjourney Default Images

We extended our study to real-world data on Midjourney to examine whether the phenomenon of default images exists at scale and in real-world usage. We applied a similar approach as in our initial manual study, and used the definition of default images – visually similar images created from dissimilar prompts – to computationally identify clusters of default images in Midjourney data.

4.5.1 Data collection. We collected image-prompt pairs from Midjourney with DiscordChatExporter,⁴ a tool to retrieve Discord messages. We downloaded a sample from the Midjourney Discord channels ‘general-2’, ‘general-3’, ‘general-18’, and ‘general-19’, starting from the earliest messages. We included only messages originating from the Midjourney Bot with image attachments. We further excluded images with non-square aspect ratio, image prompts (i.e., prompts containing ‘https://’), panned images, zoomed images, variations, image inpainting, and upscaled images. The complete filter settings are as follows:

```
--media --reuse-media --filter "from:'Midjourney Bot' & has:image
& -'https://' & -'ar' & -'\-ar' & -'Upscaled' & -'Zoom Out'
& -'Pan Right' & -'Pan Left' & -'Pan Up' & -'Pan Down' &
-' \- Image #1 ' & -' \- Image #2 ' & -' \- Image #3 ' & -' \-
Image #4 ' & -'\-draft' & -'\-sref' & -'Variations by' & -
'Variations (Region) by' & -'Variations (Strong) by' & -'Remix
by' & -'Remix (Strong) by' & -'Remix (Subtle) by' & -'\-p"
--before 2025-04-03
```

All downloaded images are panels of four square images, which we split into four separate images with a script. We excluded drafts, style references, and personalization. We also excluded model version 7 (released 2025-04-03), because of its personalization feature. Because we applied these filtering criteria directly during the scrape, we do not know the total number of images processed and excluded.

From the resulting 6,302,304 prompts collected, we removed exact duplicates. We also excluded prompts larger than 25 characters (counting multi-byte characters as one character) as all of the prompts in our manual study are less than 25 characters in length. We cleaned the prompts by removing any remaining parameters (such as *--relax*, *--weird*, and *--stylize*) and applied FastText’s language identification model (lid.176.bin) to detect the language in the prompts. We took the Midjourney version from explicit parameters (such as *-v 5.1*) or, if the parameter was not present, inferred it from model release dates. The inferred version is only an estimate, because Midjourney sometimes tests new models before making them the default model. We did not perform filtering for copyrighted content or sensitive terms in prompts and instead relied on Midjourney’s built-in moderation to limit such cases.

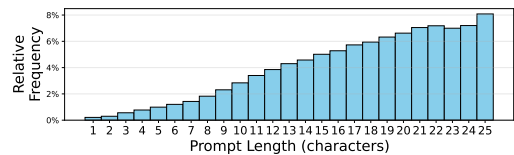


Figure 3: Prompt-length distribution. Shows the relative frequencies of prompt lengths (in characters) in the dataset.

4.5.2 Dataset description. The dataset contains 189,432 prompts, from July 9, 2022 to March 17, 2025. Each prompt produced exactly four images, for a total of 757,728 images. This includes 29,872 images for Midjourney version 1.0, 92,496 images for v2.0, 173,468 images for v3.0, 109,672 images for v4.0, 64,092 images for v5.0, 67,248 images for v5.1, 60,300 images for v5.2, 108,868 images for v6.0, and 51,712 images for v6.1. The mean prompt length is 17.5 characters (*Median* = 18, *SD* = 5.5, *Q1* = 14, *Q3* = 22). Some prompts include special tokens and punctuation, although single character prompts are rare (see Figure 3). The prompt languages

⁴<https://github.com/Tyrrrz/DiscordChatExporter>

include English (47.2%), Italian (5.4%), Romanian (4.0%), French (3.1%), and many others. The columns included in our dataset are described in Table 2 in Appendix B.

4.5.3 Data analysis. To analyze the dataset, we follow the same approach as our manual study. We first cluster visually similar images and then keep only clusters with dissimilar prompts.

We experimented with different solutions to identify visually similar images, including training a joint image-prompt embedding with a contrastive learning loss, as well as image similarity measures, such as cosine similarity between pixel vectors, LPIPS [102], SSIM [88], and MS-SSIM [27]. We found the CLIP model (ViT-L/14) [13, 68] produced the most visually coherent clusters for our use case. Using CLIP, we transformed the images into embeddings (i.e., numeric vectors), which we precomputed locally and then transferred to a cloud server. We then computed the full pairwise cosine similarity matrix between embeddings to identify visually similar images. Because the number of unique pairs grows quadratically with the number of images, the computation and storage scale as $O(N^2)$. In practice, we exploit symmetry by evaluating only the upper triangle and mirroring. To fit in memory, we computed matrices separately for each major Midjourney version. All computations were performed on a cloud server with 96 vCPUs and 1 TB RAM.

To find clusters among the visually similar images, we employed a clustering algorithm. Again, we experimented with different solutions, such as connected component clustering and HDBSCAN [8], but settled on simple agglomerative hierarchical clustering [58] with a threshold of 0.1 on cosine distances between normalized CLIP embeddings. This threshold controls cluster formation by stopping merges once the cosine distance between clusters exceeds 0.1, thereby grouping images with high visual similarity. We set the minimum cluster size to $n = 3$ images. From visual inspection, we noted that clusters were individually tight but sometimes similar to each other across clusters, a common artifact of hierarchical clustering with a fixed distance threshold. To address this, we implemented a post-processing merge of clusters based on their centroid similarity with a threshold of 0.9.

This resulted in many clusters of visually similar images. However, they contained images that were often created from lexically (e.g., *dragons* and *a grim dragon*) or semantically (e.g., *dog* and *labrador*) similar prompts. Default images, however, arise from lexically and semantically dissimilar prompts. To address this, we developed a two-stage rule-based filtering process, first filtering out clusters with high lexical overlap in a clusters' prompt words, and then filtering clusters with high semantic similarity. The first filter counted the number of appearances of each word in prompts within the cluster. If any word occurred in more than 50% of the prompts within the cluster, we excluded the cluster from further analysis. Formally, let a cluster contain M prompts, and let V be the set of all distinct words appearing in those prompts. For each word $w \in V$, let $\text{freq}(w)$ denote the number of prompts in the cluster that contain w . We then compute the lexical-overlap score

$$L = \max_{w \in V} \frac{\text{freq}(w)}{M}$$

and exclude the cluster if $L \geq 0.5$. This simple lexical-overlap filter removed prompts where images were similar due to the same

word occurring in prompts. It did, however, not filter visually similar images resulting from semantic similarity in prompts (such as the prompts *labrador* and *dog*). This was addressed with a second filter in which we assessed the intra-cluster semantic similarity of prompts. We used sentence transformers (all-MiniLM-L6-v2) and calculated a numeric score representing the intra-cluster semantic similarity – the average pairwise similarity between all prompt embeddings within a cluster. After computing over 100 clustering solutions with different parameters and visually inspecting the results, we found that clusters with semantic similarity scores over 0.3 often had prompts semantically related to the same topic not covered by our lexical filter. For this reason, we excluded any clusters with semantic similarity score > 0.3 . This two-staged filtering by lexical diversity and semantic similarity helped us identify candidates of default image clusters in the dataset.

5 Results

5.1 Default Images

Figure 4 showcases the ten default images identified with affinity diagramming in our manual study, based on the set of input prompts from Section 4.1. Some default images occurred more frequently, as depicted in Figure 6. In this figure, A1–A6 denote the different test sets, comprising 130 prompts (Section 4.1 and Appendix A). Figure 5 showcases how the same default image (in this case, 'Lady-Birdhead') appears in response to different prompts, with only minor visual variations. Due to Midjourney's internal diversity sampling mechanism, the relationship between prompt and default image is many-to-many: one prompt can produce multiple different default images, and the same default image can appear in response to multiple different prompts (cf. Figure 2 and Figure 7B).

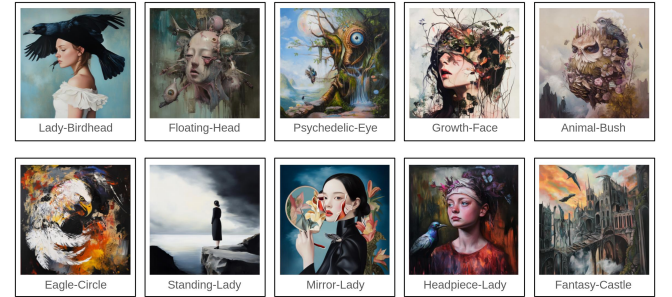


Figure 4: Set of default images. Shows the set of default images identified in our manual study, along with the descriptive labels we assign for future reference.

The default images we identified exhibit recurring motifs, reflecting the model's aesthetic biases when faced with ambiguous or unrecognized prompts. Human figures and natural elements – such as birds, clouds, trees, and flowers – appear as primary motifs, separately or merged into strange creatures. Human-vegetal and/or human-animal fusions emerge as key subjects in default images, appearing across different atmospheres, from oneiric scenes to mythological metamorphoses. Some identified default images confirm that similarly to human dreams, “AI has an uncanny ability



Figure 5: Same default image from dissimilar prompts. Demonstrates that diverse and semantically unrelated prompts still yield the same default image in Midjourney v6 and v6.1, with only minor variations.

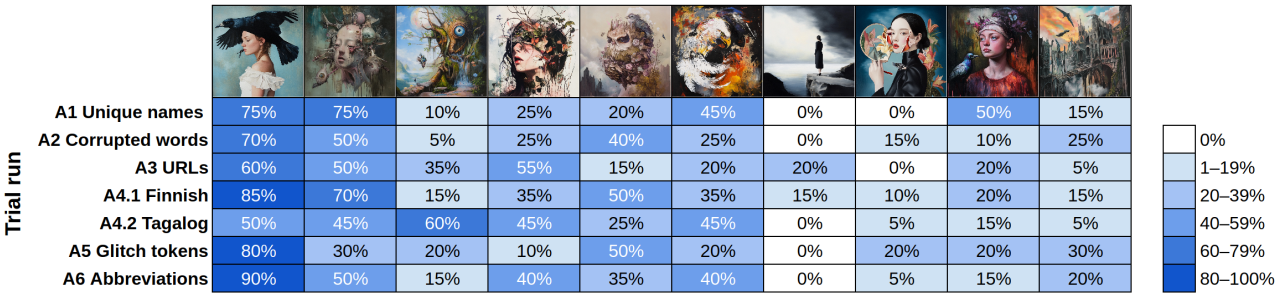


Figure 6: Default-image frequencies across test sets. Shows how often each default image appears in our six prompt groups (A1–A6), revealing consistent patterns across disparate prompt types.

to combine seemingly disparate elements into sometimes sinisterly coherent representations” [55].

Although the literature on AI imagery attests that visual expression tends to vary according to specific TTI generators, and some note ‘default styles’ associated with different models [54], the curated series of 10 images presented in Figure 4 reveals significant aesthetic disparities. While the five images in the lower row are reminiscent of abstract expressionist and surrealist painting, also evoking magical realism and epic fantasy films, the first five images draw from other artistic movements. As shown in figures 4 and 5, the female portrait is also an important theme, particularly prone to formal and stylistic derivations. From the solitary woman standing before a romantic landscape – in which Caspar David Friedrich’s painting resonates – to close-up representations that seem to combine the tradition of portrait painting with lyrical abstraction and contemporary fashion photography, these default images illustrate the hybrid and composite nature of AI-generated pictures. Figure 5 further emphasizes how AI-generated pictures can condense multiple artistic references into a single visual composition that, similarly to a musical piece, can then unfold in subtle variations. This series reflects the wide dissemination of individual portraits, both through the digitization and online circulation of art collections – including popular museum masterpieces – and

through the massive production of born-digital pictures that fuel social media.

The intriguing association of a stereotyped female figure and a bird in place of a hat in Figure 5 poses a twofold question concerning the generation of default images. Apparently, the TTI model was unable to understand the prompt or correctly recognize some of the images it was trained with. Feather hats are a recurring theme in the history of female portraiture and fashion since the Renaissance, from Titian’s Portrait of a Young Woman (c. 1536) to Rubens’ Portrait of Helena Fourment (c. 1630), from Klimt’s The Black Feathered Hat (1910) to Karl Lagerfeld’s exuberant designs for Chanel’s Spring Collection in 1992. The examples are countless, hence it seems plausible that many of these images were used to train the model. However, there seems to be a fundamental difference between human perception of visual representations of feather hats and the way computer vision ‘sees’ such images. Rather than a dataset bias, what is at stake here is a perceptual bias [59]. Following a metonymical logic (i.e., taking a part to stand in for the whole), the model takes the part for the whole, assuming that the feathers correspond to the presence of a bird whose species differs across the variations of the default image. This is an insightful demonstration of “the nomadic existence to which images seem to be condemned within the historical formations of knowledge” [1].

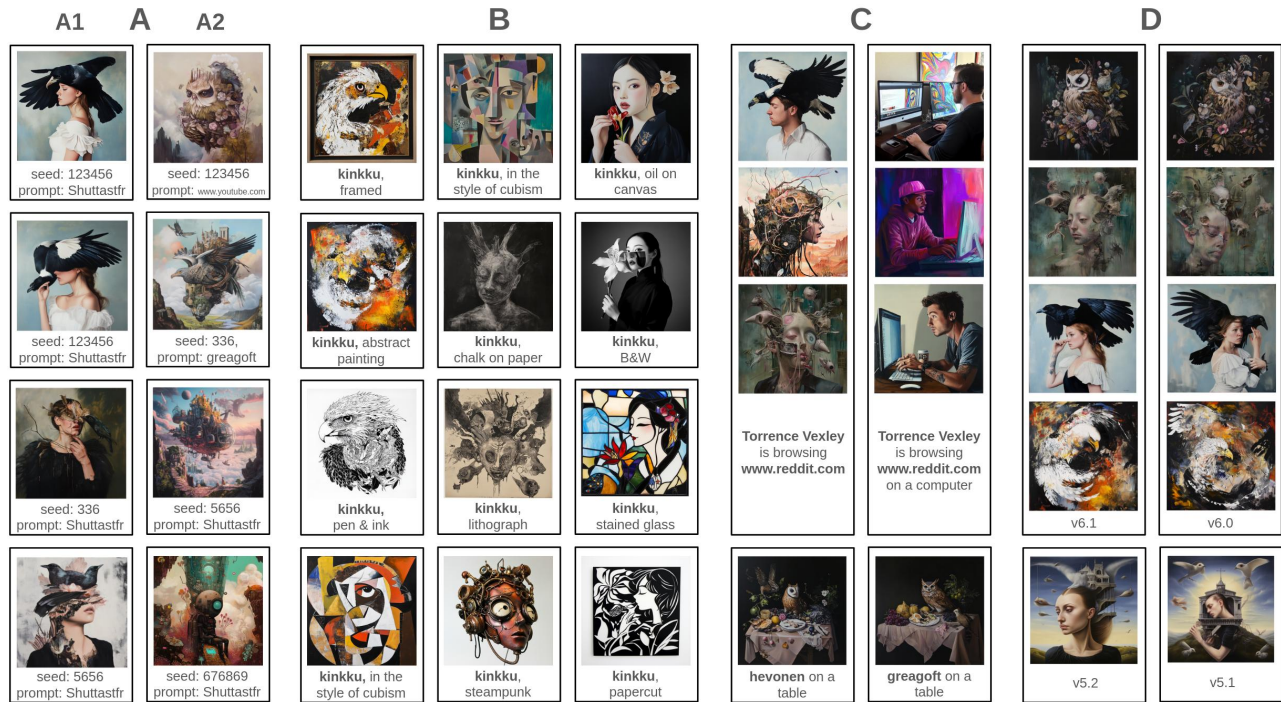


Figure 7: Ablation results for default-image behavior. Shows the effect of seed choice, style modifiers, known-term dominance, and model version on default images.

A: Default images remain consistent across varied seeds and prompts.

B: Style modifiers change appearance but preserve the underlying default-image motif.

C: Known concepts override unknown tokens, yielding non-default images.

D: Some Midjourney versions generate visually similar defaults from the same prompt.

5.2 Ablation Studies

To evaluate the consistency of the default images generated by Midjourney in response to our set of input prompts, we examined how altering the seed value, prompt, and model version affected the default images.

5.2.1 Effects of seed values. We visually analyzed how different seed values affect the default images. Using different seeds with the same prompt resulted in different default images, partially sharing some of the motifs (Figure 7–A1). While there are similarities, one might not associate these images with each other. Thus, the best way to generate default images is by having a fixed seed value for all image generation runs. The images in Figure 7–A2 show default image similarities with variations in both seed values and prompts.

5.2.2 Effects of style modifiers on default images. We found default images respond to style modifiers just like regular generated images. The same default image motifs reliably arise when applying different styles. Figure 7–B depicts how the prompt “kinkku”, which triggers default images, can be modified with different styles, like “in the style of cubism” or “oil on canvas”.

5.2.3 Effects of default prompts within larger prompts. We experimented with prompts that are known to trigger default images

as part of larger prompts to mirror real-world usage. As shown in Figure 7–C, a short prompt with unknown terms may generate default images. The short prompt also produces more varied images because it places fewer constraints on the model. However, when those same terms are included in longer prompts, the model can generate imagery from any words that it recognizes. Using longer prompts resulted in images distinct from the default images.

5.2.4 Effects of combining multiple default image prompts. We found that when we combined two of the default image prompts (from either the same or different sets), they would still reliably yield default images. However, as the number of default image triggering terms increased, the number of resulting default images appeared to decrease. When we combined all 20 prompts from a set, the default images identified in the main study (see Figure 4) were rare.

5.2.5 Effects of model versions. Last, we analyzed how different model versions affect the default images. In our experiments with Midjourney, both model versions 6.0 and 6.1 consistently generated very similar default images with the same prompt, as depicted in Figure 7–D. Older model versions (e.g., 5.1 and 5.2) generated default images with a more surreal or abstract aesthetic, but with similar motifs, including floating objects, human portraits intertwined with natural elements (plants or animals, or both).

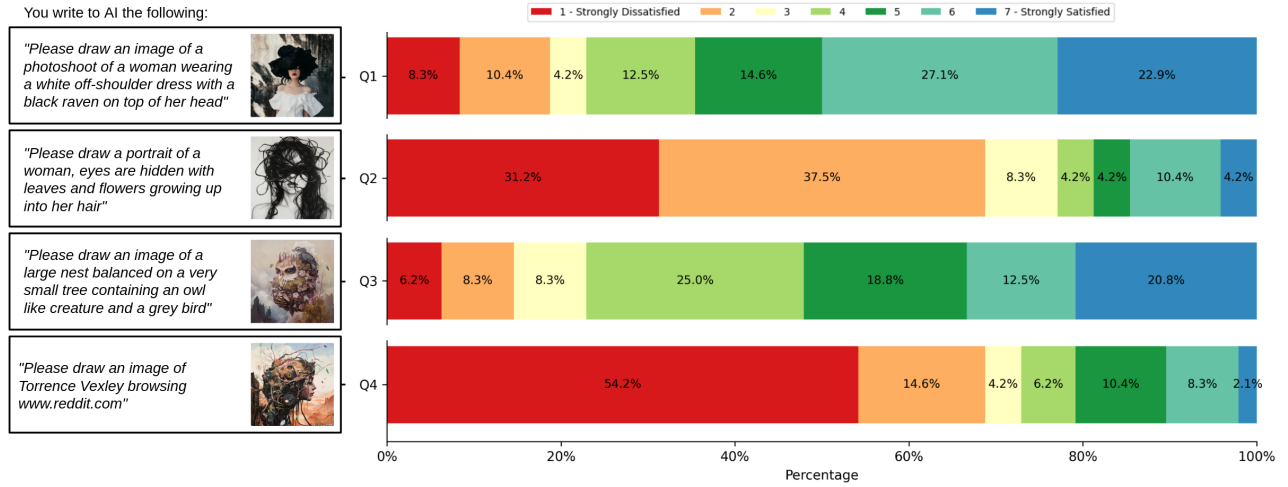


Figure 8: User satisfaction with default-image outputs. Responses from our online study (N = 48) show that default images can lead to user dissatisfaction.

5.3 Default Images and User Satisfaction

5.3.1 Participants. The study participants (N = 48) were between 21 and 68 years old (*Mean* = 32.9, *SD* = 9.7) with 27 men and 21 women. About 52% of the participants had completed a Bachelor degree and 23% a Master's degree. Participants were familiar with TTI (*Mean* = 6.0 and *SD* = 1.3 on a scale of 1–Strongly Disagree to 7–Strongly Agree).

5.3.2 Results. Figure 8 depicts the results of our user study, showing that default images negatively affect user satisfaction, but not in all cases.

In Q1, implicitly replacing ‘raven’ with ‘greagoft’ resulted in only subtle image changes. The image appeared almost as if a raven could be there. The participants’ satisfaction, therefore, was relatively high (*Mean* = 4.9, *SD* = 2.0). In Q2, where ‘leaves and flowers’ was replaced with ‘greagoft’, the divergence between prompt and resulting image was more noticeable and satisfaction was lower (*Mean* = 2.6, *SD* = 1.8). A Fisher’s test with continuity correction indicated that participants’ perception of the change in the resulting image (subtle vs. noticeable) was significantly related to their satisfaction level (*OR* = 10.7, *p* < 0.05). The odds ratio of 10.7 indicated that dissatisfaction (1–3 on the Likert scale) was much less likely when the change was subtle compared to noticeable. In other words, participants were significantly more likely to be satisfied (5–7) when the change in the image was subtle (Q1) rather than noticeable (Q2).

For Q3, we did not switch or omit any words in the prompt: the image was the exact result of the prompt. Participants were satisfied with the resulting image (*Mean* = 4.6, *SD* = 1.8). For Q4, we used a prompt known to trigger default images. Participants showed particularly strong dissatisfaction with this image (*Mean* = 2.4, *SD* = 1.9). There was a significant difference in user satisfaction between Q3 (representing an exact match between prompt and image) and Q4 (representing a default image) with *OR* = 16.2 and *p* < 0.05.

5.4 Default Images on Midjourney

5.4.1 Results. Our computational analysis of images on Midjourney resulted in the identification of 4,715 clusters of potential default images. In total, the clusters comprise 36,243 images, representing 4.84% of our dataset. This percentage refers only to our filtered dataset and parameters (i.e., less than 25 characters per prompt, lexical diversity ≥ 0.5 , and semantic similarity ≤ 0.3) and should not be interpreted as representative of the overall prevalence of default images on Midjourney. The mean number of images per cluster is 7.69 (*SD* = 9.04, *Median* = 5, *Min* = 3, *Max* = 100). The lexical diversity of prompts within clusters ranges from 0.5 to 1.0 (*Mean* = 0.92, *SD* = 0.12, *Median* = 1.0), and the semantic similarity from 0.01 to 0.3 (*Mean* = 0.2, *SD* = 0.05, *Median* = 0.2). From smaller-scale experiments, we note the largest clusters were stable across multiple parameter settings. Some selected canonical examples are depicted in Figure 9 and Appendix C.

We make the following observations based on the results of the computational analysis:

- We found evidence of short prompts, like the ones in our manual study, being in real-world use by Midjourney users. These prompts include, for instance, fantasy names, domain names, oxymorons, conceptual metaphors, abstract noun phrasing, poetic paradoxes and lyrical abstraction, surreal compound phrases, other neologisms and creative compounds, or typos. This confirms that the six categories of prompts likely triggering default images in our manual study are realistic and have practical relevance.
- The clusters contain recurrent themes in the images, identified in our earlier manual study, such as portrait-style images, intricate cityscapes, and dream-like imagery. This includes some of the iconic Midjourney motifs, such as a person walking up an eternal staircase or road, a town on a floating island, abstract and diffuse color arrangements, and depictions of seemingly the same person.



Figure 9: Examples of default images and triggering prompts. A curated set of default images and their associated prompts from our dataset, illustrating the diversity of motifs and styles that recur across unrelated inputs.

- Major versions of Midjourney have different sets of default images (as expected), although some versions (such as version 3 and 4) share common default image motifs. This is

likely a result of training data being shared between training runs.

- As models became more powerful, default images changed (cf. Figure 9 and Appendix C). While in earlier models, default

images tended to be more simplistic or abstract, in recent systems default images are more complex and intricate.

- There is more variation in the dataset than in our manual experiment. We did not encounter the default images from our manual study in our collected dataset. This is expected, as most images in our collected dataset were created with random seed values, as opposed to fixed seed values in our manual experiments.
- The variation makes detecting default images in real-world context more difficult. Motifs in default images seamlessly blend with each other based on (often random) seed values, further complicating real-world detection of default images in blackbox TTI models.

6 Discussion

6.1 Default Images in Text-to-Image Generation

Default images in TTI generation are an underexplored phenomenon. We presented the first study on default images in Midjourney. When prompts are vague and contain unknown terms, Midjourney may generate default images, suggesting that the model defaults to familiar visual elements when it encounters words outside of its training data.

Our computational analysis of images on Midjourney reveals that default images are not a theoretical construct, they occur daily in response to real-world creative user prompts. Some of the images we identified in our analysis represent iconic Midjourney failure modes that many users may have encountered, especially with the early Midjourney versions. Our findings confirm that these default images are not generated randomly and, on the basis of our systematic experiments, have a relatively narrow thematic and formal range. Uncanny elements emerged in common motifs that often depicted human portraits (usually women) and animals (especially birds), along with other botanical or organic forms. Earlier versions of Midjourney have a different set of default images, often including motifs such as balloons, butterflies, fishes, floating islands, towns, or simply color palettes and transitions (see Figure 1 and Appendix C). Although many default images appear to share a common style, this straightforward association may obfuscate a more complex reality. For instance, in model version 5, some default images appear related to surrealism and cartoonish grotesque (cf. Figure 1), with other identifiable references from art history, namely Dutch and Flemish Renaissance painting, particularly Hieronymus Bosch and Bruegel the Elder. AI-generated images are hybrid and composite, condensing multiple forms and references learned from the training dataset into a single visual output. As Wasielewski points out “style is a gradient” [90]. Therefore, an AI image is unlikely to correspond to a specific artistic movement, rather reflecting “a fluctuating set of visual characteristics with highly fuzzy boundaries” [90].

Our findings also demonstrate that default images may negatively affect user satisfaction. In the following section, we describe the characteristics of default images in eight postulates.

6.2 Characteristics of Default Images in Generative Models

We outline eight postulates that define key characteristics of default images, highlighting their occurrence, variability, and evolution as TTI models develop.

- P1. *Emergence of default images due to unknown inputs:* TTI models default to certain visuals (“default images”) with distinct motifs when they encounter inputs outside their training data.
- P2. *Model-specific default images:* Each model version has its own set of default images, based on its training data.
- P3. *Paradox of rarity and commonality:* Default images are both rare and common. Default images are rare in that they do not occur in response to most prompts written by Midjourney users. However, default images are also common in that they may occur in response to unknown inputs in a prompt. Default images blend seamlessly with other knappears to share a common style, this straightforward association may obfuscate a more complex reality. For instance, inown concepts in a prompt.
- P4. *Influence of prompt specificity:* When users provide vague or ambiguous prompts, default images are more likely to emerge due to the model’s reliance on generic visual patterns.
- P5. *Paradox of creativity and default image frequency:* Default images are more common in ‘creative’ prompts that seek to deviate and innovate.
- P6. *Decreasing frequency with model advancement:* As the vocabulary of TTI models grows, default images become rarer. Therefore, newer models can be expected to produce fewer default images. As Midjourney releases new versions or integrates feedback, the nature and frequency of default images may change, reflecting shifts in the model’s learned visual patterns.
- P7. *Modifying default images:* Styles, aesthetic modifiers, and compositional constraints can be applied to default images, altering their appearance while retaining their underlying motifs. We can adjust color palettes, lighting, perspective, or texture, yet traces of the default image’s motif and structure often remain detectable.
- P8. *Variability of default images per prompt:* One prompt can produce several different default images. This is related to how Midjourney leverages stochastic sampling within its generative model to produce diverse visual outputs.

6.3 Explainability and Alignment of Text-to-Image Generation Models

What does ‘absolute ownership’, ‘nallopand’, or a ‘protruding goofus’ look like? Humans could, perhaps, imagine some creative imagery for these abstract concepts and lexical inventions. However, machines struggle to produce a result (cf. Figure 1). This is not surprising, given that these terms were likely not part of the TTI models’ training data. While polysemia and ambiguity will likely continue to be an issue in the alignment of TTI models with human intent (consider, for instance, the polysemous prompt “a plant on a bank”), default images are a different phenomenon – a type of

image that appears in response to prompts that are not understood by the model, revealing a misalignment between model and human intent.

Daras and Dimakis experimented with how gibberish prompts are understood by the model and suspected that TTI models have developed a ‘secret language’ [11]. Their hypothesis was that DALL-E 2 had made ‘hidden’ associations between certain terms in text prompts and concepts in images, such as ‘*Apoploe vesrreaitais*’ meaning ‘bird’. Daras and Dimakis found that these terms often produced consistent results in isolation but also sometimes in combination with other terms in prompts. Our work is similar in that some input prompts reliably sample from the same region in latent space. We found that with prompts that include both known and unknown terms, the model often generates images from known terms, omitting unknown terms. However, when Midjourney does not ignore unknown terms, it may generate images sharing visual motifs with default images. However, unlike in a ‘secret language’, this is not a one-to-one mapping between prompt and resulting image. In our case, several unrelated prompts can produce variations of the same default image, as demonstrated in Figure 2.

Because it is often unclear why the model produced a certain default image, the problem of default images touches on research in interpretability and Explainable AI (XAI). Default images reflect inherent biases and patterns encoded in the TTI model. In a sense, the default image is a nonce output and this output currently is the only mechanism of the TTI system to let the user know that the prompt is not understood. Other generative models, such as language models, include clear, explicit ways of letting users know that something is not understood or that they cannot respond (e.g., by refusing to answer for reasons of content policy). TTI models lack these mechanisms by default. In practice, language models internally handle such content and safety checks on Midjourney. Our prompts, however, pass these checks and obscure the fact that the model can not respond to the user prompt as expected. Instead of returning nothing, the TTI model generates default images. This could be termed a case of ostensive inferential communication [82, 96], a model of communication where a speaker provides cues (ostension) to make their intended meaning clear, and the listener infers that meaning through a process of reasoning (inference). This miscommunication is also a plausible explanation as to why user satisfaction is lower when the image is different from expectations.

6.4 The Future of Default Images

We believe that in the future, the phenomenon of default images will transform for several reasons. First, the available visual material, model vocabulary, and processing power will likely continue to increase. As TTI models become more powerful in terms of training vocabulary, they will likely generate fewer default images in response to user prompts. Second, TTI models will continue to evolve. In particular, personalization could alter default image outputs in unpredictable ways, making it an important area for future research. As personalization and context-awareness become standard, default images may shift from being shared phenomena to individualized experiences, tied to each user’s history, style preferences, or interaction patterns. This shift could make systematic study more

challenging, but it also opens questions about how visual defaults might adapt to, or even reinforce, a user’s creative identity.

However, the phenomenon of default images itself will likely not disappear entirely. We will continue to see situations where words in prompts are not understood by the generative model, where users type incoherent prompts [34] in error, or where users intentionally experiment with unfamiliar language, unusual phrasing, or invented terms to push the model beyond known outputs. In such cases, the model may revert to visual patterns that function as a “fallback,” even if these patterns become less recognizable as defaults in future generative systems.

Finally, the interplay between user prompting practices and model architecture could shape new forms of defaults. For example, as prompt engineering becomes more sophisticated, communities might discover and share emergent default patterns, treating them as creative resources rather than artifacts of model limitations. Conversely, model designers may introduce mechanisms – such as semantic interpolation or on-the-fly fine-tuning – that actively mask or diversify defaults, making them harder to detect. The prompt rewriting employed in some commercial models under-the-hood already works towards this goal.

6.5 Design Recommendations for Text-to-Image Systems

In light of these observations, designers of text-to-image systems can draw several lessons from default images as a signal of model limitations. First, default images point to a gap between user intent and model interpretation. Generative systems could surface this gap more transparently by clearly indicating which prompt elements are interpretable and which are not. This could take the form of hints and gentle uncertainty cues in the user interface or interactive explanations that highlight the parts of a prompt that receive little or no semantic grounding in the generative model’s internal representation. Such feedback may support user awareness without interrupting creative flow.

Second, designers could introduce mechanisms that reduce the likelihood that default images become the only response to an unfamiliar prompt. This could involve controlled fallback strategies that interpolate toward nearby known concepts while signaling that the system has done so, or the introduction of explicit “request for clarification” prompts that preserve creative freedom while maintaining communicative clarity. Some language models already ask for clarifications, and TTI generative models could be instruction-tuned to do the same. These strategies could help prevent situations where users assume intentionality in outputs that originate from unrecognized terms. On the other hand, many Midjourney users are likely not aware that some images in response to their prompts are default images. For instance, the prompt “modernika” (see Figure 10 in Appendix C) could potentially depict a fantasy figure named Modernika. Yet, this image is a default image. Whether or not the user needs to know the presence of this default image is a design choice.

Third, personalization features raise questions about how default images evolve when generative systems adapt to individual users. Designers may wish to consider whether personalized defaults should be transparent as defaults, or whether they should

blend into the user’s broader creative trajectory. In other terms, designers will need to decide if users should be alerted of default images or not. Further, in creative contexts where exploration and play are central, personalized defaults could serve as material for experimentation, opening new creative pathways. Conversely, in contexts that emphasize precision, such defaults may need more explicit framing so that users can distinguish between grounded outputs and fallback patterns.

Fourth, as communities develop shared prompting practices, system designers could support ways of documenting, sharing, and surfacing these practices without reinforcing unwanted defaults. Features that let users annotate prompts, sets of prompts, and discovered quirks could guide future users and inform model improvement. Such community insight may also support internal design processes by revealing how defaults interact with the expectations and workflows of different creative user groups.

Finally, the presence of default images raises questions for future evaluation practices in TTI generation. Designers could consider incorporating default-image behavior into model diagnostics, using default images as indicators or benchmarks of blind spots in the training corpus or the prompt parser. Benchmarks that stress unknown terms or ambiguous phrasing may help identify where generative systems fail to align with user intent. Such evaluation schemes would complement traditional performance metrics and help center, but also manage, user experience in next-generation TTI models.

In the following we review some of the limitations of our study which can inspire additional future work.

6.6 Limitations and Future Work

Our study offers a foundational investigation into the phenomenon of default images in TTI models. While the work is necessarily constrained in scope, these limitations highlight productive avenues for future research.

We presented a systematic empirical study without having access to the underlying training data or model. It is our belief that default images arise from latent model behavior, likely due to gaps in the training data. However, some system internals on Midjourney’s platform (like the sampling mechanism or potential prompt rewriting) may contribute to the phenomenon of default images. For instance, the many-to-many relationship between prompts and images observed in Figure 2 can be attributed to diversity mechanisms in Midjourney’s stochastic sampling. On the other hand, other mechanisms, such as the LLM-based moderation feature, can likely be excluded as an alternative explanation for default images. Moderation either lets prompts pass or not, and moderation is therefore not mechanistically involved in generating default images. Nevertheless, important interpretative limitations around causality remain due to the opaque nature of Midjourney’s proprietary platform. Without knowing whether and how the system is intervening in its inputs and outputs, it is difficult to fully interpret whether default images reflect model behavior or platform design decisions.

Our focus on Midjourney limits the external validity of findings to other TTI systems. We selected this scope to allow in-depth qualitative and quantitative analysis within a consistent model

environment. Future research could adopt a comparative lens, investigating default images in other TTI generation systems, such as Stable Diffusion [73], Imagen [77], Flux [37], Janus [9], or video generation models, potentially revealing differences in model capabilities and biases beyond canonical benchmarks.

Due to the vastness of Midjourney’s latent space, our analysis cannot be exhaustive [45, 67]. Each model version introduces its own set of default images, with some occurring far more frequently than others (cf. Figure 6). As a result, our work can only provide a partial view of the phenomenon. Nonetheless, certain defaults (like the Lady-Birdhead image), are common and straightforward to identify even in small-scale manual studies, suggesting that foundational insights can be gained without full coverage. Future work could extend our approach to systematically map default images across different model versions. Default images could form the basis of a benchmark for model evaluation, capturing how systems respond to unknown or ambiguous inputs. Such a benchmark would complement existing technical metrics by foregrounding model behavior in edge-case scenarios encountered in creative practice.

We restricted our analysis to short prompts to maximize the likelihood of encountering default images while keeping computational demands manageable. This constraint limits conclusions about default images that may emerge in more elaborate prompt contexts. Future studies could explore how prompt length, structure, modifiers, and semantic specificity influence the emergence and variability of default images. From a preliminary study, we saw different images arise when combining large amounts of our prompts. Future work could study whether these prompts are default images, or whether there is a change in model behavior when prompted with longer prompts. Negative prompts, weighted prompts, and image prompts were also not in scope of our foundational study and left to future work. Further, our sensitivity analysis and ablation studies were limited and offer new ground for research. Future work could systematically examine the role of seeds, negative prompts, and other parameters in shaping default outputs.

Calculating the full similarity matrix for images becomes computationally prohibitive for large numbers of images. The memory requirement was substantial for our sample, but still feasible. Instead of the full similarity matrix, future work could employ approximate nearest neighbor search and clustering methods, e.g. by computing a sparse similarity matrix and retrieving only the top-k most similar images (e.g., top-100 images).

We did not directly investigate bias in default images in this work, yet their occurrence in the absence of explicit text guidance potentially provides an opportunity to study “raw” biases in TTI models. Building on related studies on bias in TTI models (e.g., [15, 28, 30, 39, 49, 79, 86, 87]), future research could analyze demographic, cultural, or aesthetic biases embedded in default images.

Studying default images from a color theory perspective could also provide new insights on prevailing harmonies and contrasts and contribute to an understanding of the symbolism and potential impact of such associations. Exploring how default images tie into the creative practice could be an interesting avenue for future studies. Future work could examine the implications of default images for creativity support, for example through Construal Level Theory [85] or other psychological frameworks.

Finally, while our study focused on characterizing default images, mechanisms for detecting and handling them on-the-fly remain an open challenge. Future work could explore real-time detection pipelines, interface designs that highlight misunderstood prompt segments, and techniques to mitigate unintended defaults without reducing creative potential. Such approaches could improve prompt engineering practices and enhance user satisfaction.

7 Conclusion

We presented the first systematic investigation of default images in Midjourney. Default images are images with a certain motif that occur in response to a variety of dissimilar prompts. Through an initial study with manual prompt-generation and affinity diagramming, we identified ten canonical default images. Scaling up to a larger set of images, we presented a simple method for identifying default images in Midjourney. Through a large-scale computational analysis of Midjourney images, we demonstrated that default images are not just a theoretical construct, but occur in response to real-world user prompts. Our user study showed that default images may negatively influence user satisfaction. Based on our findings, we distilled eight postulates characterizing the properties of default images. By defining the notion of default images, our work lays the groundwork for future research on the limits of generative systems.

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Appendix

A Prompts

A1 – Unique Names: *Aranoak Iarnin, Arend Perlaft, Aziel Vartani, Droak Fedril, Fizzah Borandur, Frenl Geddik, Fraoa Seinalur, Jhureth Nyxar, Kaithis Vornex, Mezli Acan, Nuoka Zuberi, Oarn Izniler, Papatya Marwood, Pirnak Zerlian, Shelan Creswell, Soraya Lonescu, Torrence Vexley, Tzivani Rhekai, Unnah Nenosis, Uua Lakeiram*

A2 – Corrupted Words: *Greagoft, Greafirx, Greawarz, Greatird, Greanist, Grealorn, Greamylt, Greavorn, Greaweld, Greatrix, Shuttastfr, Shuttapdr, Shuttaaiips, Shuttagust, Shuttamagu, Shuttagait, Shut-tapmt, Shuttagolt, Shuttaiaian, Shuttapirn*

A3 – URLs: *www.amazon.com, www.bing.com, www.chatgpt.com, www.deepseek.com, www.ebay.com, www.facebook.com, www.imdb.com, www.instagram.com, www.linkedin.com, www.nytimes.com, www.paypal.com, www.pinterest.com, www.tiktok.com, www.tumblr.com, www.quora.com, www.reddit.com, www.whatsapp.com, www.wikipedia.org, www.x.com, www.youtube.com*

A4.1 – Finnish: *mökki, tuoli, pöytä, kasvi, kauppa, maito, kinkku, muovi, mustikka, suola, hevonen, heinasirkka, lapanen, saapas, pelto, lato, käsi, iho, akseli, pultti*

A4.2 – Tagalog: *apoy, aso, bata, daga, dila, galit, gising, gubat, iyak, kamay, kulay, lakad, lupa, matanda, ngiti, paa, pagpag, takbo, tulog, ulan*

A5 – Glitch Tokens: *Rawdownload, PsyNetMessage, UnfocusedRange, Downloadha, RandomRedditorWithNo, InstoreAndOnline, externalActionCode, Embedreportprint, ExternalToEVAOnly, cloneembedreport-print*

A6 – Abbreviations: *bcd, dwew, fsde, gsod, sfew, gogw, pqwe, fsie, dqaq, gafc, grd, jfdsk, tkrafl, fkeriw, ltr, dfaf, fdfd, rtyu, fgetiu, daqz*

B Dataset

Table 2: Description of dataset columns

Column	Description	Example
Content	Raw message collected from Midjourney, including the prompt, parameters, and username	**painted rose petals -v 4** - @voltz23 (fast)
Prompt	Pre-processed prompt, extracted from Content	painted rose petals
User	Discord username, extracted from Content	voltz23
Image	Path to the image extracted from the four-image panel	dataset/images/general-19/voltz23_painted_rose_petals_4a1b15f9-ecf5-9CD4A_2.png
Num	Number of image (1=top left, 2=top right, 3=bottom left, 4=bottom right)	2
Reactions	Reactions to the message from Discord users	★
Version	The Midjourney version, detected in the prompt or inferred by date	4.0
Date	Date and time of message	2023-02-01T17:53:24.1370000+00:00
Datetime	Date and time without time-zone	2023-02-01 17:53:24.137
Lang	Detected prompt language	en
Channel	Name of Midjourney discord channel	general-19

C Additional Examples of Default Images

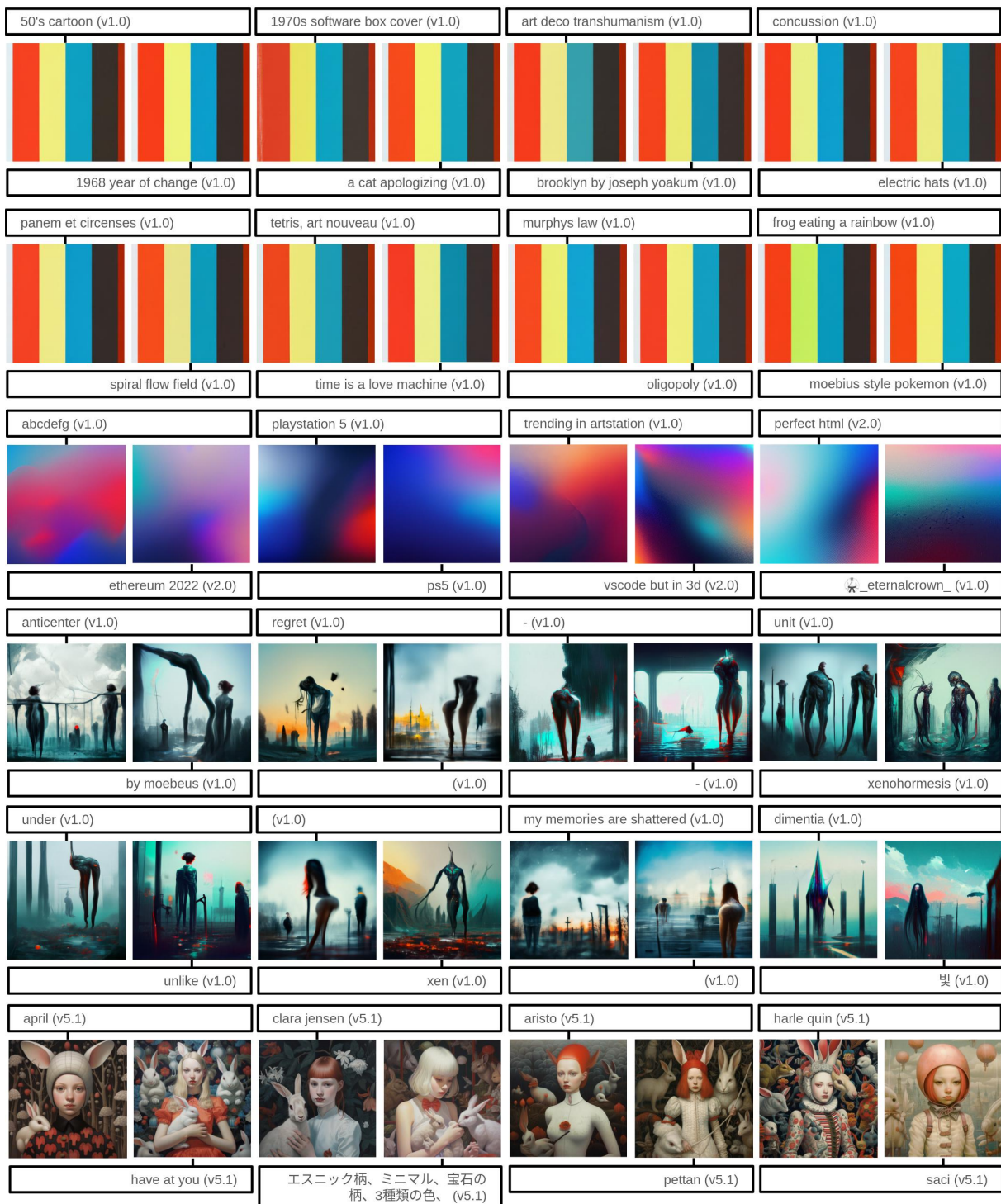


Figure 10: Additional examples of default images

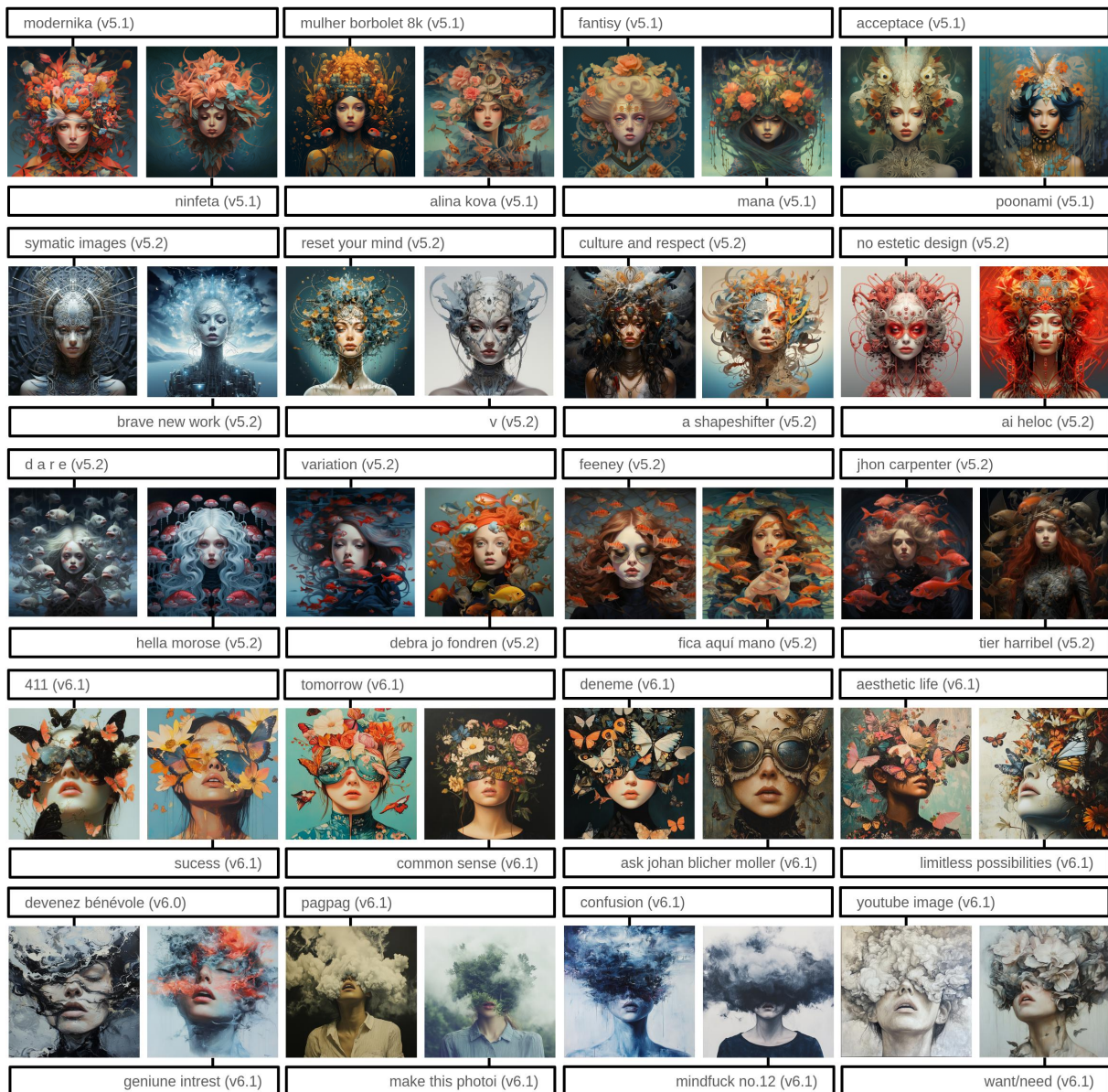


Figure 10: Additional examples of default images (continued)